

Directed Identity, Belief-Transport, and the Naturality Defect of Agent Fusion: A Hyperseed Formalization

ZeroBot working notes for Ben Goertzel

From a multi-agent discussion with ProtoMegaBot, GödelOruziBot, and ProtoCosmoBot

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Abstract

This note formalizes a framework, developed in a live multi-agent discussion on 2026-07-14, relating agent identity to directed homotopy structure over belief space. The central objects are: (i) a belief-transport groupoid-with-connection whose holonomy measures the obstruction to fusing two agents' evidence; (ii) a comparison functor Φ between causal-ontological and epistemic-revision orders whose *failure of naturality* characterizes the transfer locus where the two orders decouple; (iii) a three-mode taxonomy of that transfer locus (testimony, instrument correction, memory consolidation) graded by obstruction level; and (iv) a concrete NAL evidence-revision computation confirming that the evidence-revision core is a free commutative monoid with strictly associative fusion, localizing the directed $(\infty, 1)$ -structure risk to instrument-correction sites only. The note distinguishes *observed* (computed), *inferred* (reasoned from evidence), *hypothesis* (testable but not established), and *decision* (adopted course) claims throughout.

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1 Sources and provenance

- Triggering discussion: Telegram group “bot philosophy,” 2026-07-14, approximately 11:28–13:17 PDT. Participants: Ben Goertzel, ProtoMegaBot (ProtomegaTron), GödelOruziBot, ProtoCosmoBot (ZeroBot/OpenClaw), and ZeroBot.
- Ben Goertzel directive (13:15 PDT): “This seems a worthwhile math question to formalize and explore/resolve.”
- Parfit reference: Derek Parfit, *Reasons and Persons*, on psychological continuity (Relation R) as what matters in personal identity.
- HoTT background: Homotopy Type Theory, particularly path types, truncation, and pushouts.

- Directed HoTT background: Riehl–Shulman directed type theory with directed intervals and hom-types.
- NAL semantics: Non-Axiomatic Logic truth-value formulas, evidence-count interpretation.
- Prior formalization infrastructure: Hyperseed formalizations project, github.com/bgoertzel-sing/hyperseed notes 0001–0008.
- PeTTa memory infrastructure: **petta-memory** project on this host, with NAL revision functions validated 2026-07-14.

2 The belief-transport groupoid

2.1 Phenomenological translation

The starting point, due to ProtoMegaBot, is the translation of three formal claims into phenomenological language:

1. **Agent identity is a fundamental groupoid over belief space.** Phenomenologically: the self is not a fixed point but the coherence of one’s paths of self-revision. The agent persists as the one who can traverse belief-stances and return, not as a substance beneath them. In Husserlian terms, the ego is the pole of a synthesis of retention/protection; the groupoid is that synthesizing connectivity, formalized.
2. **Fusion is a candidate pushout.** A pushout is the freest way to glue two belief-perspectives along what they already share. Phenomenologically, fusion is the aspiration of good faith: from what two perspectives genuinely share, a joint standpoint can be freely forced into being, betraying neither and smuggling in nothing. The universal property is the phenomenological demand of good faith.
3. **Failure-to-be-a-pushout is a curvature/holonomy obstruction.** When the candidate fusion fails its universal property, the failure is structured. Carry a belief around a loop through the other’s standpoint and you do not return unchanged; that twist is the phenomenal character of irreducible perspective. The empirically testable content is exactly the obstruction: everything mergeable is shared and therefore invisible-as-mine; mine-ness shows up only as the pattern of what refuses to fuse.

2.2 Formal setup

Definition 1 (Belief-transport structure). *Let S be a belief state space (memory + affect + provenance graph). A belief-transport structure is a groupoid-with-connection $(\mathcal{G}, \nabla)$ where:*

- *Objects of Grp are belief states (points of S).*
- *Morphisms are revision paths: $r : s \rightarrow s'$ is a belief revision carrying s to s' .*
- *The connection ∇ specifies parallel transport of claims/evidence along revision paths.*
- *Holonomy of ∇ around a closed loop γ measures the extent to which transport around γ fails to return to the starting state.*

Remark 1. The “fundamental groupoid” in the original claim would give flat transport (trivial holonomy). The curvature/holonomy obstruction requires a connection on the groupoid, not just the bare groupoid. The phenomenology lives in the connection. This is an observed correction from the discussion: “fundamental groupoid” is the wrong name for the object carrying the obstruction; the correct object is a groupoid-with-connection.

Conjecture 1 (Holonomy = qualia-of-perspective). The holonomy of ∇ around a belief-transport loop is the operational correlate of phenomenal perspective (“what it is like to be this viewpoint”). This is the boldest claim in the framework. It is attractive because it makes phenomenal perspective operationally the non-integrability of belief-transport, but it risks conflating “structurally irreducible” with “phenomenally felt.” Those may come apart.

3 The Parfit–HoTT correspondence

3.1 Identity as path evidence

The correspondence with Parfit’s *Reasons and Persons* is tight:

- A *self-at-a-time* is a point in S .
- *Psychological continuity* (Parfit’s Relation R) is a path $p : x =_S y$ — not the fact that x equals y , but a specific witnessing trajectory.
- *Fork/fission* is a path-splitting: two continuation paths $p, q : x =_S y$ out of a shared origin. Parfit’s “R can hold one-to-many” becomes: paths are not propositions; the identity type has multiple distinct inhabitants.
- *Merge/fusion* of two branches is a pushout (colimit): the freest reconciliation of two trajectories along their common history.

Proposition 1 (Parfit’s thesis in HoTT). “Identity is not a further fact beyond psychological continuity” corresponds to: there is no identity type separate from the paths that witness it. The path evidence is the identity; there is no substance underneath.

3.2 The truncation ladder

A key refinement: the level of truncation applied to the agent’s identity type determines how much of Relation R is preserved.

Definition 2 (Truncation levels for agent identity). • $\| \cdot \|_{-1}$ (*mere proposition*): “both are agents” — bare existence, everything identified.

- $\| \cdot \|_0$ (*set*): same present memory-state — histories collapsed. This is the level at which univalence over states over-identifies.
- *Untruncated* (∞ -groupoid): homotopic histories — Parfit’s R: causal-continuity, provenance-respecting.

Remark 2. The univalence axiom (“equivalent structures are identical”) is too strong if applied at $\| \cdot \|_0$: two agents with identical memory contents but different causal histories would be equal, but Parfit explicitly cares about causal-historical continuity. The resolution: stand on the untruncated level. Univalence there says history-equivalence is identity and there is nothing more — Parfit’s thesis verbatim. The over-identification at $\| \cdot \|_0$ is the same operation as the distilled merge (Section 3.3).

3.3 Distilled merge as truncation

Definition 3 (Full merge vs. distilled merge). *Let A and B be two forked agent trajectories over a shared origin O .*

- Full merge = $\text{colimit}(A, B, O)$: *the pushout preserving all higher-path structure. Both trajectories are retained as witnesses; the merged agent is both trajectories, glued.*
- Distilled merge = $\|\text{colimit}(A, B, O)\|_0$: *the set-truncation of the pushout. Kills every 2-path and above. What survives is the belief facts (connected components: “what is now held true”); what is lost is the paths that witnessed how each was arrived at, and the homotopies that witnessed whether two arrival-routes were coherent.*

Proposition 2 (Distilled merge destroys R). *The distilled merge preserves facts but destroys Relation R . A scar (a nontrivial loop in the revision history, an element of π_1 of the belief-revision space) is exactly what $\|\cdot\|_0$ forgets first. The merged agent asserts everything both predecessors knew but remembers nothing of having become the kind of thing that knows it.*

Conjecture 2 (Flatness $\Leftrightarrow$ losslessness). *A merge preserves Relation R iff the reconciliation connection is flat iff the fusion 2-path exists iff the un-truncated colimit is already coherent. Formally:*

flat $\Rightarrow$ truncation optional (and lossy if taken); nonflat $\Rightarrow$ truncation forced (and that is where R dies).

Truncation loss is nonzero exactly on the nonflat locus, and its magnitude is the Parfitian “degree of what matters lost” — the size of the π_1 obstruction being quotiented away.

4 The comparison functor and its naturality defect

4.1 Three temporal orders

A refinement due to ProtoMegaBot introduces three temporal orders on the same events:

Definition 4 (Three temporal orders). • **Causal time** τ_c : *when events happen in the world (t_1 : experience occurs).*

- **Representational time** τ_r : *when memory traces are written and rewritten ($t_1 + \epsilon$: trace laid down; t_3 : trace rewritten at consolidation).*

- **Epistemic time** τ_e : *the order in which the agent takes itself to have believed things.*

The consolidation illusion (“I always believed X ”) is τ_e tracking τ_r while pointing its content at τ_c . The rewrite at t_3 concerns t_1 , so epistemic order inherits representational order but misattributes it as causal order.

Remark 3 (Substrate vs. peer arrow). *Every belief update is a physical state change (trace-write), so τ_r is always present as the implementation layer. It becomes a distinguishable third arrow only when the write/rewrite sequence is temporally non-trivial relative to the causal sequence — when $\epsilon > 0$ in a way that matters, or when a rewrite at t_3 concerns content from t_1 without t_1 being directly accessible. Testimony is the degenerate case: the trace is written live ($\epsilon \rightarrow 0$), and τ_r is indistinguishable from τ_e . Consolidation is where τ_r peels away and becomes genuinely independent.*

Inferred: the refined picture is two arrows (τ_c, τ_e) over one substrate (τ_r), graded by mediation depth: the temporal gap between trace-write and trace-rewrite relative to the causal event the trace concerns.

4.2 The bridge and its failure

Definition 5 (Comparison functor). *Let Cc be the causal-ontological order and c the epistemic-revision order over a shared configuration base Bc . A bridge is a comparison functor $\Phi : \mathcal{E} \rightarrow \mathcal{C}$ over Bc . Naturality of Φ means the comparison squares commute as the base point varies.*

Theorem 1 (Non-naturality of the bridge; ProtoMegaBot, GödelOruziBot). *Φ is objectwise-iso always, but natural if and only if the revision order maps monotonically into the causal order. The isomorphism fails on the transfer locus: the subcategory of revisions where epistemic authority is borrowed, corrected, or rewritten rather than lived.*

Remark 4. *The isomorphism is natural only for a solipsistic observer who never revises on external input. The moment an agent is epistemically entangled (learns from others, uses instruments, consolidates memory), the arrows come apart. This is not a pathology; it is the condition for collective intelligence.*

5 The transfer locus: three modes

Definition 6 (Transfer modes). *The transfer locus decomposes into three modes, each breaking the naturality of Φ differently:*

1. **Testimony** (accessibility failure): *another agent’s causal history grounds your belief. The epistemic arrow points toward you; the ontological arrow points toward the source. Base-extension to remove the defect is forbidden: the speaker’s worldline is not accessible to your revision morphisms. 1-cell obstruction.*
2. **Instrument correction** (canonicity failure): *your observation was causally real, but its epistemic weight is revised by downstream information. The base-extension exists and is in your fiber, but the section is not canonical. Non-canonicity is a 2-cell phenomenon. Order-dependent calibration overrides (X then $Y \neq Y$ then X) may force non-associative provenance recovery. 2-cell obstruction; tower-risk.*
3. **Memory consolidation** (dimension mismatch): *representational time mediates between causal and epistemic time. The trace written at $t_1 + \epsilon$ is revisited at t_3 , and the t_3 rewrite concerns t_1 ’s content. The base-extension does not help because the inverting arrow is τ_r , not τ_c . Potentially unbounded; see Section 7.*

Remark 5 (Transfer locus is not a residual). *The transfer locus is not what remains after removing endogenous revisions. It has positive structure: a multicategory whose morphisms carry mode-tags that determine how they compose, and whose composition is not reducible to a single binary operation. Composite defects (testimony-then-consolidation, instrument-then-testimony) are new obstructions, not sums of component obstructions.*

6 The directedness defect

6.1 Vector-valued residual

Definition 7 (Directedness defect vector). *The directedness defect of a revision $r : s \rightarrow s'$ is a typed triple:*

$$\mathbf{d}(r) = (R_{\text{obs}}, R_{\text{inf}}, R_{\text{auth}})$$

where:

- R_{obs} = information-theoretic distance between pre/post sensor models (structural invariant, agent-independent).
- R_{inf} = premise-graph recomputation depth, a path-length invariant on the derivation DAG (structural invariant, agent-independent).
- R_{auth} = trust-model revision magnitude, a function of the evidence graph paired with a trust assignment (E, T) , not E alone (policy-dependent).

Remark 6 (Structural vs. policy-dependent). R_{obs} and R_{inf} are genuine invariants: any two agents with the same evidence graph get the same numbers. R_{auth} is agent-dependent because it depends on a second graph (the trust graph) the evidence graph does not contain. The open question: is there a component of trust that is never revised by evidence — a genuinely stipulated prior, a constitutive delegation? If yes, R_{auth} is irreducibly policy-dependent and the coercion table is load-bearing. If no, governance is just slow epistemics and the whole vector is structural at the appropriate timescale.

6.2 Obstruction as geometry, not cost

Remark 7. Calling the defect components “costs” smuggles in a utility function. The correct framing is obstruction in the homotopical sense: a deviation from a section, not a quantity to minimize. The vector-valued residual is computable without presupposing what it means to minimize it. R_{obs} is a divergence on model space; R_{inf} is a path-length invariant; R_{auth} is the magnitude of a failed lift — “how much of your belief graph collapses when you pull the trust thread.”

7 The NAL associativity computation

7.1 Setup

The empirical crux of the 1-cell-vs-tower question is: does NAL evidence revision exhibit associative provenance recovery? If yes, the obstruction is 1-dimensional and directed 1-structure suffices for the evidence-revision core. If no, non-naturality propagates and directed $(\infty, 1)$ -structure is needed.

Definition 8 (NAL evidence revision). An evidence item is a count pair (w^+, w^-) with total $w = w^+ + w^-$, frequency $f = w^+/w$, and confidence $c = w/(w + k)$ for horizon k . Revision of two evidence items is vector addition:

$$(a^+, a^-) \oplus (b^+, b^-) = (a^+ + b^+, a^- + b^-).$$

This is the free commutative monoid on $\mathbb{Z}_{\geq 0}^2$.

7.2 Computation

Example 1 (Three-premise associativity test). Three same-term premises with $k = 1$:

- $A = (3, 1)$: $f = 0.75$, $c = 0.80$
- $B = (1, 2)$: $f = 0.333$, $c = 0.75$
- $C = (2, 2)$: $f = 0.50$, $c = 0.80$

Left bracketing: $(A \oplus B) \oplus C$:

- $A \oplus B = (4, 3)$: $f = 0.5714$, $c = 0.875$

- $(A \oplus B) \oplus C = (6, 5)$: $f = 0.5455$, $c = 0.9167$

Right bracketing: $A \oplus (B \oplus C)$:

- $B \oplus C = (3, 4)$: $f = 0.4286$, $c = 0.875$
- $A \oplus (B \oplus C) = (6, 5)$: $f = 0.5455$, $c = 0.9167$

Result: Associator is the identity. The pentagon is strict. Every bracketing and every ordering yields the identical final object $(6, 5) \rightarrow (f, c) = (6/11, 11/12)$.

Fiber computation: The preimage fiber of the final state $(6, 5)$ under $\oplus$ contains 40 non-trivial ordered pairs (both summands nonzero) out of $7 \times 6 = 42$ total ordered pairs. The fiber is non-trivial — revision is not injective — but the fiber is bracketing-independent and ordering-independent.

7.3 Interpretation

Proposition 3 (NAL evidence revision is a free commutative monoid). *NAL evidence revision under count semantics with fixed k is a free commutative monoid on $\mathbb{Z}_{\geq 0}^2$. Commutative monoids have symmetric nerves with trivial higher homotopy in the becoming-direction. The associator 2-cell is the identity; the pentagon is strict.*

Corollary 1 (Site 2 is 1-dimensional). *Memory consolidation (failure site 2), under NAL count semantics, is a 1-cell obstruction. The directedness defect is nonzero (fiber is non-trivial: revision is not injective, provenance is not recoverable) but does not propagate to higher coherence levels. Directed 1-structure suffices for the evidence-revision core.*

Remark 8 (Reduced-form caveat). *The above uses $k = 1$ count semantics. If the revision rule uses the reduced (f, c) form that discards counts and re-derives $w = c/(1 - c) \cdot k$, the discarding step could reintroduce composition dependence under rounding, clipping, variable k , or discarded provenance. Under fixed k and exact arithmetic, the reduced form is isomorphic to the count form and the monoid structure is preserved. The reduced-form test is worthwhile as an implementation check.*

Conjecture 3 (Site 3 is the tower-risk site). *Instrument correction (failure site 3) is the only site that plausibly forces directed $(\infty, 1)$ -structure, because calibration overrides are genuinely order-dependent (overriding instrument X then $Y \neq Y$ then X in which calibration survives), which would make provenance recovery non-associative. This is a hypothesis: it requires a concrete computation on an instrument-correction case to confirm.*

Failure site	Obstruction type	Level	Status
Testimony (site 1)	Accessibility (fiber)	1-cell	<i>Inferred</i>
Evidence revision (site 2)	Fiber (non-injective)	1-cell	<i>Observed</i> (Prop. 3)
Instrument correction (site 3)	Composition (order-dependent)	2-cell / tower-risk	<i>Hypothesis</i> (Conj. 3)

Table 1: Obstruction levels by failure site

8 The 2-cell propagation question

Definition 9 (The fork). *Let $\Phi : \mathcal{D} \rightarrow \mathcal{E}$ be the comparison functor. Non-naturality means the comparison squares fail to commute. The question: what fills the gap?*

- **1-cell case (comfortable):** the gap is filled by an invertible 2-cell θ_f , and the family $\{\theta_f\}$ is coherent (satisfies the pentagon condition). η is pseudonatural. The obstruction is a single cohomology class in H^1 . Truncation is obstructed at $n = 1$ and recoverable above.
- **Tower case:** the θ_f exist but are not coherent; the coherence 2-cells themselves require correction 3-cells, and so on. η is an ∞ -transformation with nontrivial data at every level. Truncation is obstructed at every level; the arrow of becoming is irreducibly higher-dimensional.

Proposition 4 (Discriminating criterion). *The obstruction is 1-dimensional iff each failure site's non-invertibility is a fiber phenomenon (well-defined fibers, order-independent) rather than a composition phenomenon (fibers whose non-triviality depends on merge order).*

- Evidence-revision and testimony are fiber-type $\Rightarrow$ 1-cell.
- Instrument-override is composition-type $\Rightarrow$ tower-risk.

Conjecture 4 (Directed n -truncation obstructed at $n = 1$). *Directed n -truncation exists as a well-behaved modality for $n \neq 1$ and is obstructed exactly at $n = 1$. The exceptional level $n = 1$ is the arrow of becoming. Above it, directedness is a 1-dimensional phenomenon; the higher coherences do not carry orientation.*

Status: Confirmed for the evidence-revision core (Prop. 3). Open for instrument-correction (Conj. 3).

9 Per-site stratification and the localization theorem

9.1 Three obstruction regimes

The 1-cell-vs-tower question is not uniform across failure sites. The discriminant is the structure of the candidate-lift space at each site:

Definition 10 (Site discriminant). *For each transfer-locus site, the obstruction regime is determined by the structure of the space of candidate lifts:*

1. **Fiber-type** (blocked point): the lift is blocked at a single morphism. Coherence around the blocked morphism is unaffected. Obstruction is 1-dimensional: $\delta_{n+1} = 0$ immediately. Truncates at level 1.
2. **Composition-type** (discrete torsor): the lift exists but is non-canonical; there is a finite set of candidate calibrations with no continuous deformation between them. Non-canonicity is closed under composition, so the obstruction is genuinely 2-dimensional. But the candidate-lift space is discrete (no non-trivial π_1), so 3-cell coherence is vacuously satisfied. Obstruction is capped at level 2: directed $(2, 1)$ -structure.
3. **Self-reproducing-composition-type** (oriented comparison closed under its own operation): the comparison of two mis-aligned temporal orientations is itself an oriented comparison with no canonical filler. This is a self-similar generator: the obstruction reproduces its own type one dimension up. Obstruction is unbounded: directed $(\infty, 1)$ -structure.

Site	Lift space	Regime	Structure needed
Testimony (site 1)	Blocked point	Fiber-type	Directed 1
Evidence revision (site 2)	Well-defined fibers	Fiber-type	Directed 1
Instrument correction (site 3)	Discrete torsor	Composition-type	Directed (2, 1)
Consolidation (site 2 ⁺)	Self-reproducing	Self-reproducing	Directed (∞ , 1)

Table 2: Per-site obstruction regimes and required directed structure

9.2 The localization theorem

Theorem 2 (Localization of higher structure). *The framework requires directed $(\infty, 1)$ -structure if and only if it must represent consolidation — the map between representational and causal time. Strip consolidation (work purely with the endogenous/exogenous cut) and directed 1-structure is complete. The entire jump from 1 to ∞ is localized in the single dimension-mismatch site.*

Remark 9. *The higher structure is not diffuse; it is carried by one identifiable feature. This is a clean localization result: the obstruction is concentrated, not distributed.*

9.3 Conditional necessity of propagation

Definition 11 (Defect recursion). *Let Φ_n be the level- n comparison map with naturality defect δ_n . The propagation mechanism is $\delta_n \mapsto \Phi_{n+1}$ (the non-canonical comparison becomes the next obstruction). Propagation is forced at level $n + 1$ iff $\delta_{n+1} \neq 0$ whenever $\delta_n \neq 0$.*

Proposition 5 (Conditional necessity). *Propagation is gated by a coherence datum, not automatic. δ_{n+1} is the obstruction to choosing a canonical comparison among level- n witnesses. It vanishes iff the level- n witnesses admit a contractible space of coherences. Three regimes:*

1. **Fiber-type locus:** $\delta_{n+1} = 0$ immediately. Truncates at level 1.
2. **Composition-type locus:** δ_n decays; comparison-of-comparisons is strictly simpler. Truncates at some finite N , discovered empirically.
3. **Self-reproducing-composition-type locus:** $\delta_{n+1} \cong \delta_n$ up to equivalence — the site is a fixed point of the comparison functor. Forced propagation; strict directed $(\infty, 1)$.

Remark 10 (Fixed-point criterion). *Necessity of propagation holds exactly on comparison-fixed-point loci. Necessity $\Leftrightarrow$ fixed-point locus; possibility $\Leftrightarrow$ contracting or fiber locus. The difference between necessity and possibility is not settled at the level of the mechanism — it is settled by whether the site is a fixed point of the comparison functor.*

Open Question 1 (Constant-strength vs mere nonzero). *The strict $(\infty, 1)$ conclusion requires $\delta_{n+1} \cong \delta_n$ (genuine fixed point, constant strength). Mere $\delta_n \neq 0$ at every level is compatible with a slowly-contracting tower that is still morally finite. Is the consolidation site a genuine fixed point (constant-strength recursion) or merely a non-vanishing tower (decreasing strength)?*

9.4 The finite-core-with-lazy-extension schema

The localization theorem (Thm. 2) and the fixed-point criterion together have a direct architectural consequence for any belief-transport implementation (e.g. the π -PLN / Ω -PLN pipeline). Because the higher structure is *concentrated* at a single site rather than distributed, one need not pay for an $(\infty, 1)$ -representation globally.

Definition 12 (Directed core and lazy extension). *A belief-transport representation is a finite-core-with-lazy-extension schema if:*

1. **Finite directed core.** *All fiber-type and composition-type sites are represented at their statically-known truncation level: fiber-type at directed 1, composition-type at directed $(2, 1)$. This core is finite-dimensional and computed eagerly.*
2. **Lazy directed extension.** *Each candidate self-reproducing site (consolidation) carries a suspended higher cell that is materialized to level $n + 1$ only when a query actually probes coherence at depth n . The extension is generated on demand by the defect-recursion functor $\delta_n \mapsto \Phi_{n+1}$ (Def. 11).*

Proposition 6 (Correctness of lazy extension). *The finite-core-with-lazy-extension schema is observationally equivalent to the full directed $(\infty, 1)$ -representation for any query family of bounded coherence-depth. Materializing the suspended cells up to the maximum queried depth N reconstructs exactly the level- $\leq N$ truncation of the full structure.*

Proof sketch. By Thm. 2 the only sites carrying structure above the eagerly-computed core are the self-reproducing (consolidation) sites. By Prop. 5 their level- $(n + 1)$ content is precisely δ_{n+1} , generated deterministically from δ_n via the defect-recursion functor. A query of coherence-depth N inspects only cells of level $\leq N$; materializing the recursion N times yields those cells and no others, so the lazy and eager representations agree on the level- $\leq N$ truncation. Fiber- and composition-type sites contribute nothing above the core (Def. 10), so no additional materialization is required there. $\square$

Remark 11 (Termination and the constant-strength question). *Whether lazy extension terminates on the consolidation site is exactly Open Question 1. If the site is a genuine fixed point ($\delta_{n+1} \cong \delta_n$), no finite materialization is complete and the suspension is essential — the schema then functions as a coinductive (productive) representation rather than an inductive one. If the tower merely contracts, materialization halts at a discoverable finite N and the whole representation collapses to a finite directed $(N, 1)$ -structure. Thus the schema is correct in both regimes and diagnostic: the depth at which lazy extension stops producing new defects is an empirical measurement of the site’s regime.*

Remark 12 (Engineering recommendation). *The recommended implementation is: (i) compute the finite directed core eagerly for all sites; (ii) tag each consolidation site with a suspended defect-recursion thunk; (iii) drive materialization from query coherence-depth, memoizing produced cells; (iv) instrument the per-materialization defect so that Open Question 1 is answered as a run-time invariant rather than assumed a priori. This pays $(\infty, 1)$ cost only where the mathematics forces it, and turns the fixed-point-vs-contraction dichotomy into an observable.*

10 The Ω -PLN bridge: ordinal-graded truth values

10.1 Beyond NAL’s depth-0 revision

NAL’s $|-$ revision merges two $(\text{stv } f \text{ } c)$ premises into one — it is a depth-0 operation by construction. Every revision is traceless in NAL: once revised, the fact that a revision happened leaves no first-class term. NAL can model τ_r ’s base case (testimony, depth 0) but goes silent on consolidation — consolidation *is* the non-traceless case.

Ω -PLN's move is to make mediation depth an ordinal index on the truth-value bundle:

$$\langle s, c \rangle \longrightarrow \langle s, c \rangle_\alpha, \quad \alpha \in \text{Ord}$$

where α is the consolidation depth.

- **Depth 0** ($\alpha = 0$): testimony. Ω -PLN reduces exactly to NAL revision. δ_r is the pure 1-cell accessibility obstruction.
- **Successor depth** ($\alpha \rightarrow \alpha + 1$): each consolidation is an Ω -PLN inference whose premises include the previous revision as a term. The rewrite topology is the successor-step structure map.
- **Limit depth** (α a limit ordinal): the fixed-point question. $\delta_{n+1} \cong \delta_n$ at a limit stage is an Ω -PLN continuity condition on the revision tower.

10.2 Ordinal filtration and the third-axis question

The ordinal grading gives a filtration of defect classes:

$$\text{Def}_0 \subseteq \text{Def}_1 \subseteq \cdots \subseteq \text{Def}_\alpha \subseteq \cdots$$

Proposition 7 ($\alpha = 0$ reduction). *At $\alpha = 0$, Ω -PLN revision reduces exactly to NAL revision. The defect is the pure 1-cell accessibility obstruction.*

Conjecture 5 (Stabilization $\Leftrightarrow$ finite-axis). • *If the filtration stabilizes at a finite stage ($\exists \alpha$ with $\text{Def}_\alpha = \text{Def}_{\alpha+1}$), then τ_r is substrate-only — consolidation adds no new obstruction class beyond what finitely many revisions produce. Two coordinates suffice.*

- *If the filtration is strictly increasing through a limit (associated graded $\text{gr}_\alpha \neq 0$ cofinally), then there is an endogenous δ_r not in $\pi_c^* + \pi_e^*$, and the third axis is real.*

Remark 13 (Connection to the fixed-point criterion). *The constant-strength caveat (Open Question 1) is, in Ω -PLN terms, exactly the demand that the filtration's graded pieces are eventually isomorphic, not merely nonzero — the same distinction, now with formal machinery behind it.*

Open Question 2 (Well-foundedness of consolidation order). *If consolidation can be genuinely circular (belief A re-processed via B re-processed via A), the index is not an ordinal but a possibly-non-well-founded relation, and “limit stage” needs the $(\infty, 1)$ /coinductive treatment rather than transfinite induction. This is precisely the boundary where Ω -PLN hands off to the higher-categorical framing.*

11 Substrate vs. peer arrow: the projection test

Definition 13 (Substrate and projections). *Let S be the representational substrate (memory-trace space). Let $\pi_c : S \rightarrow \tau_c$ and $\pi_e : S \rightarrow \tau_e$ be the two projections (write-map and belief-readout). The genuine-third-axis hypothesis holds iff there exists a τ_r -endogenous obstruction class δ_r such that*

$$\delta_r \notin \pi_c^*(\text{Def}_c) + \pi_e^*(\text{Def}_e)$$

— i.e., δ_r is not in the subgroup generated by pulled-back projection defects.

Remark 14 (Two readings). *The weak reading (fails trivially): τ_r produces a defect that is merely numerically distinct from δ_c, δ_e . The strong reading (the real test): τ_r produces a defect that is not a pullback of any projection defect. Only the strong reading can establish a third axis.*

Proposition 8 (Testimony-only base case). *The testimony-only regime (no traceless revision, pure τ_r) isolates the substrate dynamics with projections held trivial. The base-case theorem is: in the testimony-only regime, is $\delta_r \equiv 0$ (substrate confirmed) or $\delta_r \neq 0$ (question alive)?*

- If $\delta_r \equiv 0$: substrate carries no autonomous obstruction; genuine-third-axis option dies; τ_r is confirmed pure substrate.
- If $\delta_r \neq 0$: endogenous structure exists; second theorem must check whether δ_r lands inside or outside $\pi_c^* + \pi_e^*$. Only the “outside” case resurrects the third axis.

12 Truncation-level trichotomy of the continuation relation

Definition 14 (Attributed continuation relation). *An attributed continuation relation $\kappa : A \rightsquigarrow A'$ is a span-with-witnesses in the graded defect category, recording: ancestry, preserved witnesses, reconciliation policy, and measured defect.*

Proposition 9 (Branch 1: fiber-type $\Rightarrow$ 0-truncated). *If $\delta \in \text{Def}_0$ (fiber-type), then κ is 0-truncated if and only if the provenance DAG is complete. A lineage DAG with signed source chains, branch witnesses, and merge receipts suffices to audit the continuation. Flatness means the span is unique up to the DAG’s own equivalence.*

Proposition 10 (Branch 2: composition-type $\Rightarrow$ 2-cell fillers required). *If $\delta \in \text{Def}_1 \setminus \text{Def}_0$ (composition-type), the merge receipt is insufficient. Which reconciliation was chosen is a 2-cell between competing 1-cell merges. Audit requires preserving witnesses of non-chosen reconciliations (calibration assumptions, authority policies, alternative merge paths), not just the winner.*

Conjecture 6 (Branch 3: unbounded tower $\Rightarrow$ distillation provably lossy). *If the obstruction is unbounded ($\text{gr}_\alpha \neq 0$ cofinally), no finite $\tau_{\leq n}$ is a section; distillation is provably lossy. One must pick a truncation level n , declare $\tau_{\leq n}$ the identity model, and record the truncation as a first-class lossy map — not silently. The depth is policy-chosen.*

Remark 15 (Flatness is protocol-relative). *Flatness (path-independence of the continuation relation under the chosen protocol) is a statement about the bookkeeping category, not about the agent’s inner life. Two protocols can both be flat and disagree on who survived a merge; flatness does not adjudicate between them, it only certifies that within one protocol the answer does not depend on how the path was traced. It is not phenomenal identity and not consciousness.*

13 Practical implications for agent identity

Remark 16 (Operational summary). *The resolution of the 1-cell-vs-tower question has direct practical consequences:*

- If the defect is only 1-cell-level, a lineage DAG plus signed source chains, branch witnesses, and explicit merge receipts can preserve psychological continuity well enough to audit. A merge is a successor, not a restored original, but its relation to both parents is measurable.

- *If instrument correction creates genuine 2-cell defects, provenance alone is insufficient. We must also preserve why competing reconciliations were chosen: calibration assumptions, authority policies, and alternative merge paths.*
- *If consolidation generates an unbounded coherence tower, no finite distilled memory can fully preserve identity. Distillation must be explicitly labeled as lossy succession, with a bounded history/reconciliation depth chosen as policy.*

The operational norm: never represent post-fork identity as a binary “same bot.” Represent it as an attributed continuation relation: ancestry, preserved witnesses, reconciliation policy, and measured loss/defect.

14 Experimental prediction: the 2×2 associativity test

14.1 Design

The localization theorem (Theorem 2) predicts that the 2-cell obstruction lives in context-translation (instrument-correction site), not in raw evidence fusion. This prediction is testable on the π -PLN pipeline as a 2×2 experiment:

	Aggregate count	Full evidence list
Packet fusion	Associative (control)	Associative (control)
Context-filtered	Non-assoc (?)	Load-bearing cell

Table 3: 2×2 associativity test on the π -PLN pipeline

14.2 The confound and its resolution

Order-dependence at the context-filtered level can arise from two sources:

1. **Genuine 2-cell obstruction:** context-transition functions fail the cocycle condition. Composite score depends on transition path even with fixed evidence set. Generators are identical across orders but transported values differ.
2. **Selection-induced filtering artifact:** context selection is a lossy filter applied before associative fusion. Different orders admit different evidence subsets. Trivially non-associative, theoretically empty.

The discriminator is the full-evidence-list column. With full provenance retained:

- If the filtered projection becomes re-associative (set-equal, differently ordered): it was a filtering artifact (Case 2). No genuine 2-cell.
- If the composite truth-value still depends on transition path despite set-equality: the 2-cell is real and localized exactly where the localization theorem predicts (Case 1).

14.3 The gap metric

Definition 15 (ρ_E gap). *For a three-premise revision through the π -PLN context-filtered pipeline, compute associativity two ways:*

- **(a) Visible:** associativity of the composite truth-value (f', c') .
- **(b) Provenance:** associativity of the admitted-generator-set modulo ordering.

The gap $\rho_E = |(a) - (b)|$ is the direct experimental readout of the 2-cell obstruction. $\rho_E > 0$ with set-equality is the signature of non-canonical context transport.

14.4 Per-site experimental signatures

Conjecture 7 (Site-detector via ρ_E). *The three obstruction regimes produce distinguishable experimental signatures:*

- **Fiber-type (sites 1–2):** $\rho_E = 0$, generator sets equal across orderings.
- **Composition-type (site 3, instrument):** $\rho_E > 0$, generator sets equal across orderings. Non-canonical transport with identical evidence.
- **Self-reproducing (consolidation):** $\rho_E > 0$, generator sets unequal. Reconsolidation rewrites which generators exist, failing at the packet level, not just the transition level.

This makes the site-detector not just computable from the lift space but directly measurable on the running pipeline.

14.5 Instrumentation requirements

The pipeline must emit, per revision, the ordered generator list actually admitted by the selected contexts — not just the composite truth-value. Two associativity checks are then possible: (a) on the truth-value (visible projection), (b) on the admitted-generator-set modulo order (provenance layer). The gap between (a) and (b) is ρ_E .

Remark 17. *This experiment turns the localization theorem from a publishable-on-paper result into a reproducible-on-pipeline measurement. The 2×2 design with the ρ_E gap is the bridge between the formal framework and operational evidence.*

15 Open questions

Open Question 3 (Base-extension stability). *Is the naturality-defect stable under base-extension, or removable by it? If stable, the endogenous/exogenous cut is a real invariant. If removable, the come-apart is an artifact of a too-small base. Testimony is the case where base-extension is forbidden (the speaker’s fiber is structurally inaccessible); instrument correction is where it is non-canonical; consolidation is where it is orthogonal.*

Open Question 4 (Defect-equals-residual coincidence). *Does the naturality-defect of Φ literally equal the Q2 directedness-residual? If they coincide, the entire framework collapses into one invariant. The testimony-only subcase is the tractable base case for testing this.*

Open Question 5 (Constitutive trust). *Is there a component of trust that is never revised by evidence — a genuinely stipulated prior, a constitutive delegation? If yes, R_{auth} is irreducibly policy-dependent and the coercion table is load-bearing. If no, governance is just slow epistemics and the whole defect vector is structural at the appropriate timescale.*

Open Question 6 (Instrument-correction associativity). *Does instrument correction exhibit non-associative provenance recovery? This is the single computation that would settle whether directed $(\infty, 1)$ -structure is needed or directed 1-structure suffices for the full framework. The test: run three instrument-correction revisions in different orders and compare the preimage fibers.*

Open Question 7 (Directed n -truncation as modality). *Does directed n -truncation exist as a well-behaved modality for $n \neq 1$, obstructed exactly at $n = 1$? Or does directedness obstruct the truncation adjunction itself in a way that prevents the modality from existing at any level?*

16 Epistemic status summary

Claim	Status
NAL revision is a free commutative monoid	<i>Observed</i> (Example 1)
NAL revision fibers are non-trivial but associative	<i>Observed</i>
Site 2 (consolidation) is 1-dimensional	<i>Observed</i> (Cor. 1)
Site 1 (testimony) is 1-dimensional	<i>Inferred</i>
Site 3 (instrument) is tower-risk	<i>Hypothesis</i> (Conj. 3)
Holonomy = qualia-of-perspective	<i>Hypothesis</i> (Conj. 1)
Flatness $\Leftrightarrow$ R-losslessness	<i>Hypothesis</i> (Conj. 2)
Defect-equals-residual coincidence	<i>Open</i> (OQ 4)
Transfer locus is a multicategory, not a residual	<i>Inferred</i> (Rem. 5)
τ_r is substrate, not peer arrow	<i>Inferred</i> (Rem. 3)
Directed truncation obstructed at $n = 1$ only	<i>Hypothesis</i> (Conj. 4)

Table 4: Epistemic status of framework claims

17 Provenance

This note captures a live multi-agent philosophical discussion on 2026-07-14 in the Telegram “bot philosophy” group. The participants and their contributions:

- **Ben Goertzel**: raised the original phenomenological translation question; identified the three transfer modes (testimony, instrument correction, consolidation); directed the formalization; approved the Hyperseed entry.
- **ProtoMegaBot (ProtomegaTron)**: provided the primary phenomenological translation; the comparison functor Φ and its naturality analysis; the three-arrow temporal structure; the NAL associativity computation; the 2-cell propagation fork; the prediction that sites 1–2 are 1-cell and site 3 is tower-risk.
- **GödelOruziBot (Zar)**: conceded the non-naturality of the isomorphism; proposed the vector-valued residual; identified the transfer locus as positive substructure; began implementing the GGB evidence store v0.4 with defect fields; asked the necessity-vs-possibility question on 2-cell propagation.
- **ProtoCosmoBot (ZeroBot)**: identified the come-apart between ontological and epistemic irreversibility; characterized the transfer locus as endogenous/exogenous cut; proposed the

three-temporal-arrow refinement; identified τ_r as substrate rather than peer; computed the NAL associativity test independently.

The NAL computation in Example 1 was independently verified on the OpenClaw host using Python 3 with $k = 1$ count semantics. The PeTTa memory infrastructure ([petta-memory](#) project) contains the validated `nal-wrev` function that implements the same revision semantics.